

Dan DeGenaro

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Education

- Georgetown University**, PhD in Computer Science Aug 2025 – Present
- GPA: 4.0/4.0
 - **Coursework:** Speech Processing, Seminar in NLP, Comp. Ling. Research Methods, Algorithms, Seminars in Cog. Sci., Seminar in AI Safety, Systems Programming, LLMs, Unsupervised Learning, Comp. Neuroscience
- Georgetown University**, MS in Computational Linguistics Aug 2023 – May 2025
- GPA: 3.967/4.0
 - **Coursework:** NLP, Multilingual NLP, Machine Learning, Deep Learning, Hypothesis Testing, Corpus Linguistics, Historical Linguistics, Databases
- University of Massachusetts Amherst**, BS in Physics Sep 2019 – May 2023
- GPA: 4.0/4.0
 - **Coursework:** Computational Physics, Modern Physics, Thermodynamics, Classical Mechanics, Statistical Physics, Electricity and Magnetism, Quantum Mechanics I-II
- University of Massachusetts Amherst**, BS in Applied Mathematics Sep 2019 – May 2023
- GPA: 4.0/4.0
 - **Coursework:** Calculus I-III, Linear Algebra I-II, Probability and Statistics I, Differential Equations, Chaos Theory, Numerical Analysis, Linear Optimization
- University of Massachusetts Amherst**, BA in Linguistics Sep 2019 – May 2023
- GPA: 4.0/4.0
 - **Coursework:** Phonetics, Phonology, Syntax I-II, Semantics, Sociolinguistics, Computational Linguistics I-II, NLP
 - **Minor in CS:** Algorithm Design and Analysis, Discrete Math, Data Structures
 - **Minor in Russian:** Central Asian Politics, Putin's Russia (Smith College), Russian I-IV

Research Experience

- Visiting Graduate Scholar**, Johns Hopkins University – Baltimore, MD Jun – Aug 2026
- Participated in the Human Language Technology Center of Excellence SCALE 2026 workshop.
 - Worked on extracting superior keyframes for the purposes of video retrieval.
- Visiting Research Scholar**, Johns Hopkins University – Baltimore, MD Jun – Aug 2024
- Participated in the Human Language Technology Center of Excellence SCALE 2024 workshop.
 - Contributed to a multimodal information retrieval system designed to retrieve relevant videos given text queries.
 - Developed a novel technique using downstream retrieval systems to produce preference rankings. Fine-tuned LLM using reinforcement learning to produce more retrievable document summaries.
- Undergraduate Researcher**, University of Colorado, Colorado Springs – Colorado May – Aug 2022
- Developed a novel technique for the distillation of a multilingual BERT model into a smaller model.
 - Developed original dataset, trained and fine-tuned a series of neural networks using PyTorch.
 - Wrote and presented a research paper documenting methodology and results.

Teaching Experience

- Graduate Teaching Fellow**, Georgetown University – Washington, DC Sep 2023 – Present
- 1. Fall 2026: COSC 1015 (Science of AI), undergraduate course taught by Prof. Sarah Bargal
 - 2. Spring 2026: COSC 5510 (Databases), graduate course taught by Prof. Ophir Frieder

3. Fall 2025: COSC 3470 (Deep Learning), undergraduate course taught by Prof. Sarah Bargal
 4. Spring 2025: COSC 3470 (Deep Learning), undergraduate course taught by Prof. Sarah Bargal
 5. Fall 2024: COSC 5455 (Deep Learning), graduate course taught by Prof. Sarah Bargal
 6. Spring 2024: COSC 5470 (Deep Learning for Comp. Vis.), graduate course taught by Prof. Sarah Bargal
 7. Fall 2023: COSC 5580 (Introduction to Data Analytics), graduate course taught by Prof. Lisa Singh
- Graded repository-based data science, deep learning, and database projects and associated reports and presentations.
 - Developed, debugged, tested, and graded assignments in the form of Python notebooks.
 - Designed a group project using a generative audio model.
 - Guest-lectured on PyTorch, Google Colab fundamentals, MySQL, and introductory Python programming.

Project Course Instructor, MIT MITES Semester Program – Remote Jun – Aug 202{4,5,6}

- Designed and implemented, from scratch, a machine learning and NLP course for sophisticated high school students.
- Taught advanced machine learning concepts including deep learning architectures such as transformers to ~18 students each summer for 7 weeks.
- Guided students to produce group projects using cutting-edge machine learning techniques.
- Re-hired twice (Summer 2025, 2026).

Publications

- DeGenaro, D.**, Li, X., Amo, O., Pokojovy, M., Bargal, S., Lange-Hegermann, M., & Raită, B. (2026). Fast Reconstruction of Exact Maxwell Dynamics from Sparse Data. *[SPOTLIGHT] Advances in Neural Information Processing Systems*.
- Rafkin, E., **DeGenaro, D.**, & Yang, X. (2026). Task Arithmetic with Support Languages for Low-Resource ASR. *Proceedings of the 2025 Shared Task on Low-Resource Multilingual ASR for Spontaneous Speech*.
- Abdalla, A., Shaheen, I., **DeGenaro, D.**, Mallick, R., Raita, B., & Bargal, S. A. (2025). Gift: Gradient-aware immunization of diffusion models against malicious fine-tuning with safe concepts retention. <https://arxiv.org/abs/2507.13598>
- DeGenaro, D.**, Yang, E., Etter, D., Carpenter, C., Sanders, K., Martin, A., Murray, K., & Kriz, R. (2025, August). FORTIFY: Generative Model Fine-tuning with ORPO for ReTrieval Expansion of InFormal NoisY Text. In R. Kriz & K. Murray (Eds.), *Proceedings of the 1st Workshop on Multimodal Augmented Generation via Multimodal Retrieval (MAGMaR 2025)* (pp. 100–115). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.magmar-1.13>
- Samuel, S., **DeGenaro, D.**, Guallar-Blasco, J., Sanders, K., Eisape, S., Reddy, A., Martin, A., Yates, A., Yang, E., Carpenter, C., Etter, D., Kayi, E., Wiesner, M., Murray, K., & Kriz, R. (2025). MMMORRF: Multimodal Multilingual MODularized Reciprocal Rank Fusion. *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 4004–4009. <https://doi.org/10.1145/3726302.3730157>
- DeGenaro, D.**, & Lupicki, T. (2024). Experiments in Mamba Sequence Modeling and NLLB-200 Fine-Tuning for Low Resource Multilingual Machine Translation. *Proceedings of the 4th Workshop on Natural Language Processing for Indigenous Languages of the Americas (AmericasNLP 2024)*, 188–194. <https://doi.org/10.18653/v1/2024.americasnlp-1.22>

Non-archival Conference Talks

- DeGenaro, D.** (2023). *Acting different(-ly): Bringing derivational morphology into variationist linguistics* [New Ways of Analyzing Variation].

Invited Talks and Guest Lectures

Guest lecture, Georgetown COSC 1015 30 Sep 2026

- Gave a lecture to students taking COSC 1015: Science of AI (Undergraduate) at Georgetown University.
- Demonstrated basic usage of the Python programming language.

- GUCS Research Presentation** 29 Sep 2026
- Delivered a 15-minute talk about my work, Fast Reconstruction of Exact Maxwell Dynamics from Sparse Data (DeGenaro et al., 2026) to Georgetown's CS department.
- Georgetown Computational Linguistics Joint Lab Meeting Presentation** 3 Sep 2026
- Delivered a presentation to members of all three computational linguistics labs at Georgetown University regarding usage of computing resources.
- GUCS Research Presentation** 9 Apr 2026
- Delivered a 15-minute talk about my work, Task Arithmetic with Support Languages for Low-Resource ASR (Rafkin et al., 2026) to Georgetown's CS department.
- Guest lecture, Georgetown COSC 5510** 26 Feb 2025
- Gave a lecture to students taking COSC 5510: Databases (Graduate) at Georgetown University.
 - Demonstrated basic usage of MySQL from the command line.
- Georgetown Computational Linguistics Joint Lab Meeting Presentation** 27 Jan 2025
- Delivered a presentation to members of all three computational linguistics labs at Georgetown University regarding usage of a new computing resource.
- GLSA Second Workshop on Python** 5 Dec 2025
- Delivered a follow-up 2-hour interactive session on basic use of popular Python libraries.
- GLSA First Workshop on Python** 14 Nov 2025
- Delivered a 2-hour interactive introduction to Python for absolute beginners.
- GUCS Research Presentation** 30 Sep 2025
- Delivered a 15-minute talk about my work, FORTIFY (DeGenaro et al., 2025) to Georgetown's CS department.
- Guest lecture, Georgetown COSC 3470** 18 Sep 2025
- Gave a lecture to students taking COSC 3470: Deep Learning (Undergraduate) at Georgetown University.
 - Demonstrated basic usage of PyTorch in the Google Colab computing environment.
- Guest demo, Georgetown InLab** 11 Apr 2025
- Gave a demonstration of my Whisper-UI software to fellow graduate students.
 - This demo took place at a regular meeting of the Georgetown Linguistics Department Interaction Lab, a meeting of students and faculty focused on discourse analysis.
- Virtual lecture delivered at Walter Payton College Preparatory High School** 21 Mar 2025
- Spoke to a group of high school students interested in linguistics.
 - Lectured on n -grams and provided empirical demonstrations in Python.
- Guest lecture, Georgetown COSC 3470** 6 Feb 2025
- Gave a lecture to students taking COSC 3470: Deep Learning (Undergraduate) at Georgetown University.
 - Demonstrated the training process for a simple convolutional neural network in PyTorch.
- Guest demo, Georgetown Variation Lab** 4 Feb 2025
- Gave a demonstration of my Whisper-UI software to fellow graduate students.
 - This demo took place at a regular meeting of the Georgetown Linguistics Department Variation Lab, a meeting of students and faculty focused on sociolinguistics.
- Guest lecture, Georgetown COSC 3470** 30 Jan 2025
- Same lecture as 18 Sep 2025.
- GLSA Workshop on $\LaTeX$** 20 Sep 2024
- Led a workshop for linguistics graduate students at Georgetown University.
 - Explained basic $\LaTeX$ usage, including some specialized content geared towards writing linguistics papers.
- Guest lecture, Georgetown COSC 5470** 18 Sep 2024

- Same lecture as 18 Sep 2025.

GLSA Workshop on ~~ETX~~

29 Sep 2023

- Same lecture as 20 Sep 2024.

Advising

Zehra Demirtoka and Vivi Li, Address-Level Seismic Risk Assessment from Street-View Imagery. 2026

- Student project I advised, presented at MIT's Undergraduate Research Technology Conference (URTC) as a paper.
- Fine-tuned computer vision models to assess various aspects of seismic risks given images of buildings.
- Developed a functioning web-based demo.

Ali Sahi, Abdelwahab Rahama, Micah Lynn, and Nathan Ankrah, Mechanistic Multiplicity: Implications for Circuit-level Interpretability and Intervention Effectiveness. 2026

- Student project I advised, presented at URTC as a paper.
- Trained a variety of neural architectures to examine the interplay between hidden dimensionality and neural circuit redundancy.

Ann Sali Feutchi, Megan Jeong, Court v. Chat: A Mixed-Methods Evaluation of Legal NLP Systems 2026

- Student project I advised, presented at URTC as a poster.
- Evaluated a variety of open-weight LLMs on selected tasks from LegalBench.

Keven Amaya-Muñoz, Arko Barua, and Luan Hoang, Comparative Evaluation of Domain Adaptation in Vision Models for Brain Tumor Classification With Explainability. 2025

- Student project I advised, presented at URTC as a poster.
- Benchmarked pretrained and from-scratch computer vision systems on a tumor identification task.
- Qualitatively compared explainability results and quantitatively compared strength of out-of-domain transfer.

Melissa Alfaro-Zeledon and Rida Karim, Identifying Key Factors for Femicide Prevention and Policy Development: Leveraging Supervised Machine Learning with Temporal and Geospatial Analysis 2024

- Student project I advised, presented at URTC as a lightning talk.
- Used XGBoost to detect serial offenses in a database of femicides in the United States.

Hubert Hsu and Bethany Ray, MT-MOE: Protein-Specific Drug Design Utilizing Mixture of Experts Transformers 2024

- Student project I advised, presented at URTC as a poster.
- Performance gains in drug design observed by replacing traditional transformer architecture with MoE.

Awards

NeurIPS 2026 Spotlight Award 2024-2025

- Awarded by NeurIPS Program Committee.
- Recognition for our paper Fast Reconstruction of Exact Maxwell Dynamics from Sparse Data (**DeGenaro et al.**, 2026), which was considered to be in the top 1% of all submissions, or roughly the top 5% of accepted submissions.

Graduate Student Teaching Assistant Award (Sciences) 2024-2025

- Awarded by Georgetown University.
- This award recognizes excellence among graduate students serving as TAs. Awards are given to one student from each area: humanities, social science and science and an at-large award.

Summa Cum Laude 2023

- Awarded by University of Massachusetts Amherst.
- Awarded for academic excellence (GPA in top 5%).

Commonwealth Honors College, greatest distinction 2023

- Awarded by University of Massachusetts Amherst.
- Awarded for academic excellence (completed undergraduate thesis with GPA > 3.8).

LeRoy F. Cook, Jr. Memorial Scholarship 2022

- Awarded by Department of Physics, University of Massachusetts Amherst.
- Awarded for academic excellence in physics and for engaging in teaching/tutoring as an undergraduate.

UMass Chancellor's Award 2019 – 2023

- Awarded by Admissions Office, University of Massachusetts Amherst.
- Awarded for academic excellence prior to enrollment.

Thomas J. Watson Memorial Scholarship 2019 – 2023

- Awarded by IBM Thomas J. Watson Foundation.
- Awarded for academic excellence.

Dean's List 2019 – 2023

- Awarded by University of Massachusetts Amherst.
- Awarded for academic excellence (GPA > 3.5).

Projects

FLASH-MAX ddeggenaro.github.io/flash-max

- Implementation of our technique as described in **DeGenaro** et al., 2026.

Whisper-UI pypi.org/project/whisper-ui

- GUI for OpenAI's open-source Whisper ASR systems.
- Drag-and-drop interface intended for use by linguists with little coding background.

T_EX Table pypi.org/project/tex-table

- Simple Python library to automate the formatting of L^AT_EX tables, including marking *p*-value significance.

Volunteering and Service

Treasurer, Georgetown Alliance of Graduate Employees Fall 2026 - Spring 2027

- Served as the treasurer of GAGE, a chapter of the American Federation of Teachers (AFT), a nationwide labor union.
- Represented most graduate student employees of Georgetown University.
- Responsible for the finances of the chapter (e.g. disbursing aid funds, ensuring bills are paid, etc).
- Introduced automation to some bureaucratic tasks to increase union's efficiency.

Graduate Student Node Administrator Spring 2026 - Fall 2026

- Authored extensive documentation for usage of a computing resource at Georgetown University.
- Developed useful bash scripts for users.
- Conducted extensive testing of the resource to ensure consistency and usability.
- Handled paperwork relating to access to the resource.
- Work done on a volunteer basis.

Activities Coordinator, Georgetown University Computation and Language Group (GUCL) 2025-2026

- Organized social events for graduate students/faculty in linguistics and computer science.

Treasurer, Georgetown Graduate Linguistics Student Association 2024-2025

- Organized student-led workshops and professional/networking events.

- Managed organization's sources of funding, reimbursements, catering/supplies for events, etc.

Reviewing

EMNLP Multilingual Representation Learning (MRL) Workshop 2026

ACM Multimedia (ACMMM) 2025, 2026

Hard Skills

Programming languages (Ordered by proficiency): Python, R, MySQL, C, Java, MATLAB, HTML/CSS, JavaScript, C++

Libraries (Ordered by proficiency): PyTorch (including TorchVision and TorchAudio), Jupyter, Pandas, HuggingFace Transformers, SciKit-Learn, XGBoost, NumPy, SciPy, Matplotlib, SentencePiece, Plotly, spaCy, StatsModels, Librosa, fastText, TRL, Seaborn, NLTK, Gensim, Keras, TensorFlow, Stanza, Diffusers, PyTerrier, OpenCV

Technologies (Ordered by proficiency): VSCode, Overleaf/L^AT_EX, Zotero, Git/GitHub, Slurm, Windows, Linux, RStudio, Docker, MacOS, Praat, ELAN, AWS EC2

Instructional Software (Ordered by proficiency): Google Classroom, Canvas, Moodle, Blackboard

Languages (Ordered by proficiency): Spanish, French, Russian, German